EXPERIMENT 3

DESIGNING A MINIMUM ERROR RATE CLASSIFIER

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Contents

Introduction 2
Objective of the Experiment 2
Performing the experiment 3
Task 1,2: Classifying given sample points having different colored markers according to the assigned class label 4
Task 3: Changing parameter values (μ_i, Σ_i) to see different recognition results . . 5
Task 4: Label the regions Ri for each of those two classes 7
Result Analysis 9
Conclusion 9

INTRODUCTION

The minimum error rate classifier seeks a decision rule that minimizes the probability of error which is the error rate.

For general case with risks we have,

$$g_i(x) = -R(\alpha_i|x) \tag{1}$$

because then the maximum discriminant function will correspond to the minimum conditional risk. However, for minimum error rate classifier we take,

$$g_i(x) = P(\omega_i|x) \tag{2}$$

so that the maximum discriminant function corresponds to maximum posterior probability. But we know that,

$$P(\omega_i|x) \propto P(x|\omega i)P(\omega_i) \tag{3}$$

Thus from (2) and (3) we have,

$$g_i(x) = P(x|\omega i)P(\omega_i) \tag{4}$$

Since this question involves Gaussian distribution we apply natural logrithm and get,

$$g_i(x) = lnP(x|\omega i) + lnP(\omega_i) \tag{5}$$

Equation (5) gives the discriminant function for minimum error rate classifier. Since here $P(x|\omega i)$ is a multivariate normal, that is - $P(x|\omega i \sim N(\mu_i, \Sigma_i))$ then from (5) we get,

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^T \Sigma_i^{-1}(x-\mu_i) - \frac{d}{2}ln2\pi - \frac{1}{2}ln|\Sigma_i| + lnP(\omega_i) \tag{6}$$

Since the given problem is a two class problem, we have d=2 and hence the final equation for discriminant function will be given as -

$$g_i(x) = -\frac{1}{2}(x-\mu_i)^T \Sigma_i^{-1}(x-\mu_i) - ln2\pi - \frac{1}{2}ln|\Sigma_i| + lnP(\omega_i) \tag{7}$$

And the decision boundary would be the solution of g1(x) = g2(x).

OBJECTIVE OF THE EXPERIMENT

The objective of the experiment is to design a minimum error rate classifier for a two class problem with given data :

$P(x|\omega 1) = N(\mu_1, \Sigma_1)$ $\quad$ $P(x|\omega 2) = N(\mu_2, \Sigma_2)$

$where\ \mu_1 = [0\ \ 0]$ $\quad$ $where\ \mu_2 = [2\ \ 2]$

$and\ \Sigma_1 = [.25\ \ .3\ \ ;.3\ \ 1]$ $\quad$ $and \Sigma_2 = [.5\ \ 0\ \ ;0\ \ .5]$

$P(\omega_1) = 0.5$ $\quad$ $P(\omega_2) = 0.5$

1. Classify the following unknown samples: (1,1), (1,-1), (4,5) (-2,2.5), (0,2), (2,-3)

2. Classified samples should have different colored markers according to the assigned class label.

3. Change the parameter values (μ_i, Σ_i) to see different recognition results.

4. Label the regions Ri for each of those two classes.

PERFORMING THE EXPERIMENT

Task 1,2: Classifying given sample points having different colored markers according to the assigned class label

Matlab Code 1.1

```
x1 = -7:.2:7; x2 = -7:.2:7;
[X1,X2] = meshgrid(x1,x2);
mu1 = [0 0]; Sigma1 = [.25 .3; .3 1];
F1 = mvnpdf([X1(:) X2(:)],mu1,Sigma1);
F1 = reshape(F1,length(x2),length(x1));
meshc(X1,X2,F1);
axis([-7 7 -7 7 -1.0 .6])
xlabel('x1'); ylabel('x2'); zlabel('Probability Density');
hold on;

mu2 = [2 2];Sigma2 = [.5 .0; 0 .5];
F2 = mvnpdf([X1(:) X2(:)],mu2,Sigma2);
F2 = reshape(F2,length(x2),length(x1));
meshc(X1,X2,F2);
axis([-7 7 -7 7 -1.0 .6])
xlabel('x1'); ylabel('x2'); zlabel('Probability Density');
caxis([min(F2(:))-.5*range(F2(:)),max(F2(:))]);

x = [1 1; 1 -1; 4 5; -2 2.5; 0 2; 2 -3];
for n=1:6
g1=-log(2*pi)-0.5*log(det(Sigma1))-0.5*(x(n,:)'-mu1')'*inv(Sigma1)*(x(n,:)'-mu1')+log(0.5);
g2 = -log(2*pi)-0.5*log(det(Sigma2))-0.5*(x(n,:)'-mu2')'*inv(Sigma2)*(x(n,:)'-mu2')+log(0.5);
 if g1>g2
  plot3(x(n,1),x(n,2),-1.0,'rd');
 else
  plot3(x(n,1),x(n,2),-1.0,'gs');
 end
end
```

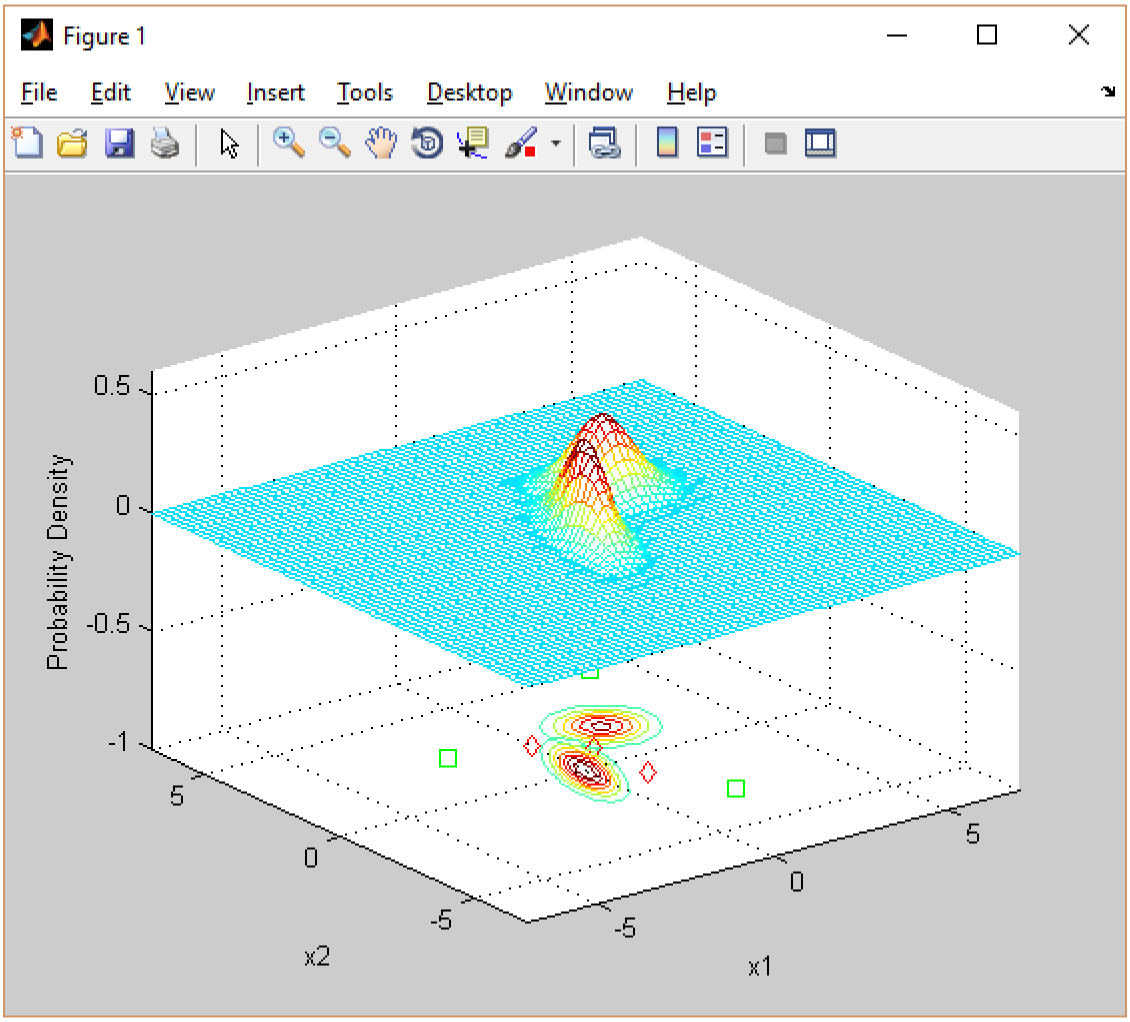

Figure 1: Output after running Matlab code 1.1

Task 3: Changing parameter values (μ_i, Σ_i) to see different recognition results

Now we will change the parameter values (μ_i, Σ_i) and later in Result Analysis section we will see the effects of the changes. We change (μ_i, Σ_i) in three ways:

- We change the value of μ_i
- We change the value of Σ_i
- We change both the values of μ_i and Σ_i

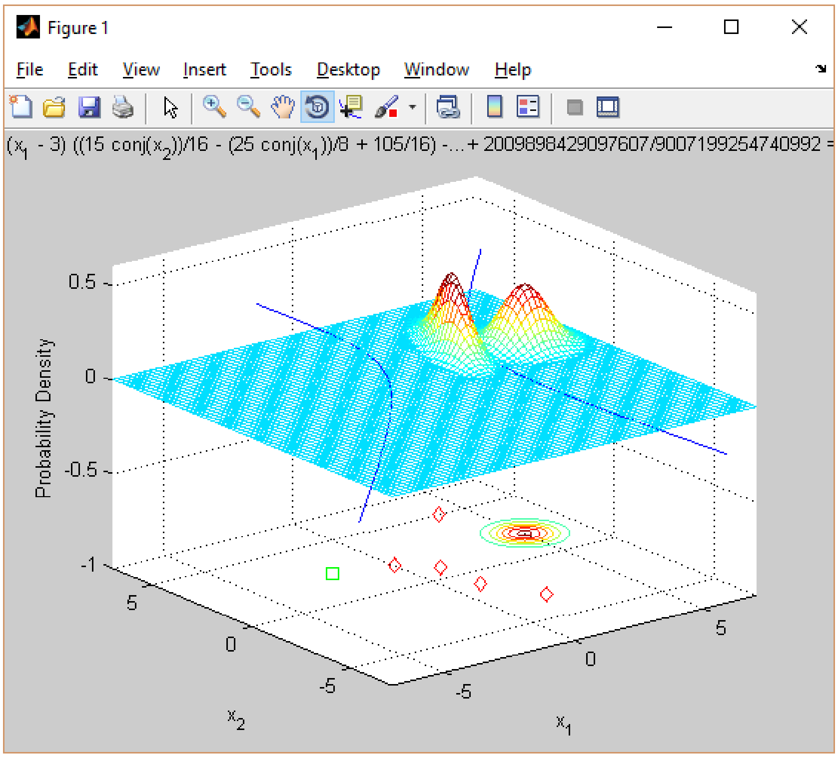

Figure 2: Case 1: mu1 = [3 3]; mu2 = [-1 2]

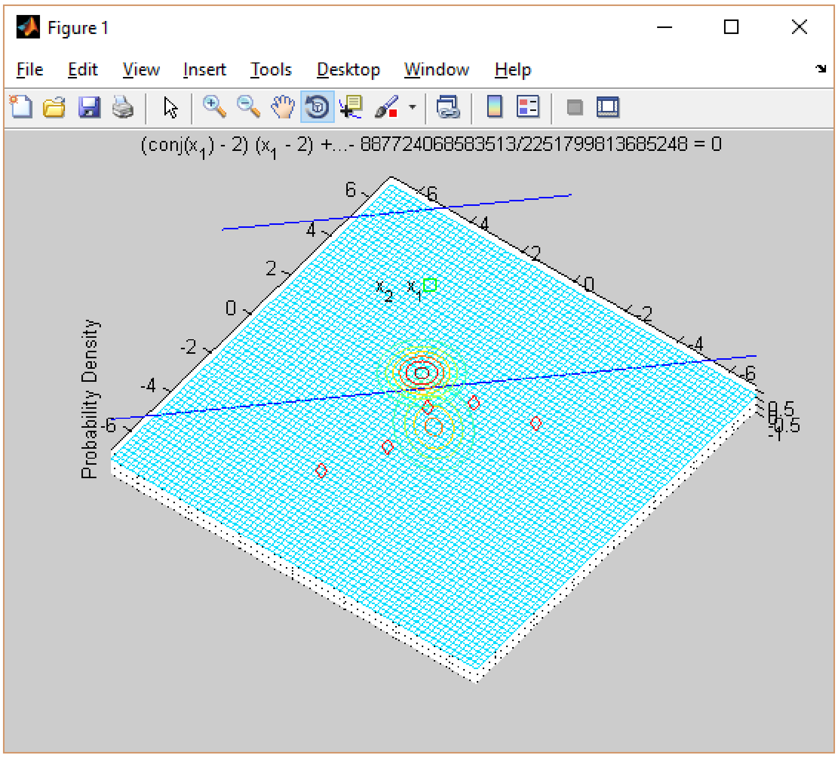

Figure 3: Case 2: Sigma1 = [.8 .3; .3 .8]

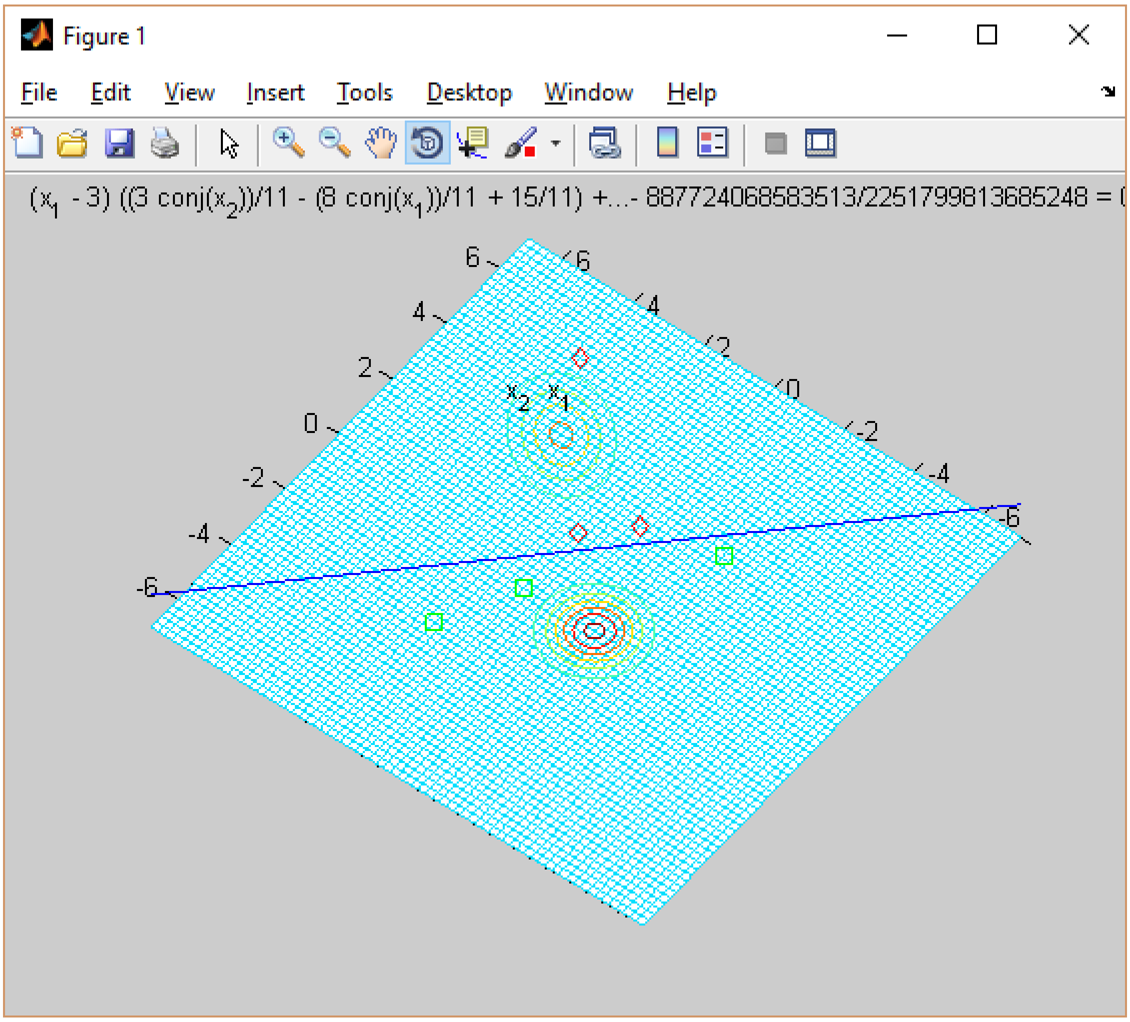

Figure 4: Case 3: mu2 = [-1 -1];Sigma2 = [.5 .0; 0 .5]; mu1 = [3 3]; Sigma1 = [.8 .3; .3 .8]

Task 4: Label the regions for each of those two classes

Matlab Code 1.3

```
syms x1 x2;
g1=-log(2*pi)-0.5*log(det(Sigma1))-0.5*(x(n,:)'-mu1')'*inv(Sigma1)*(x(n,:)'-mu1')+log(0.5);
g2=-log(2*pi)-0.5*log(det(Sigma2))-0.5*(x(n,:)'-mu2')'*inv(Sigma2)*(x(n,:)'-mu2')+log(0.5);
g = g1 - g2;
L=ezplot(g, [ [-10,10], [-10,10] ]);
```

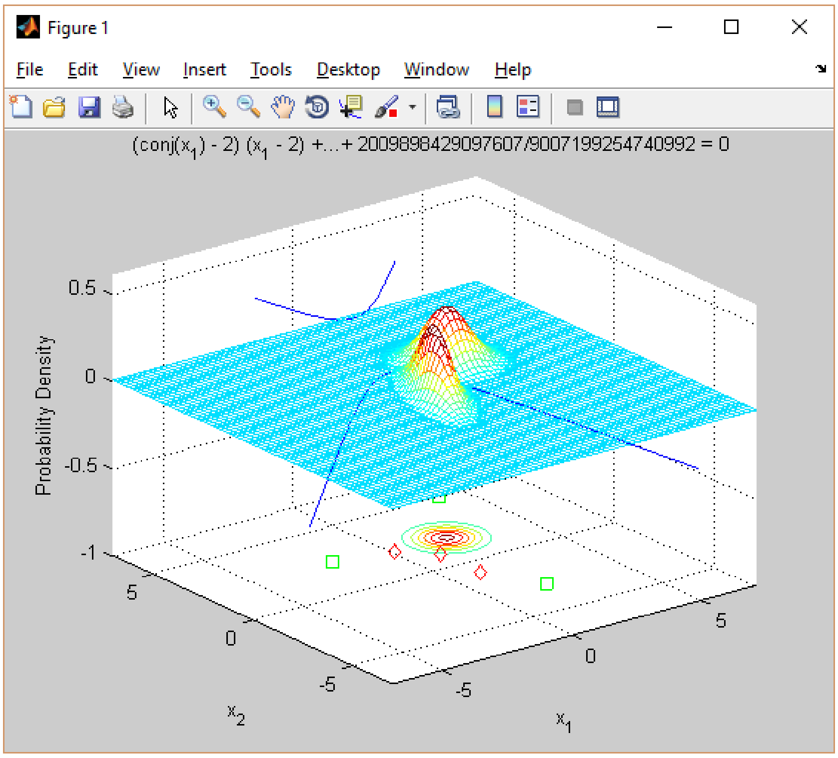

Figure 5: Output after running Matlab code 1.3

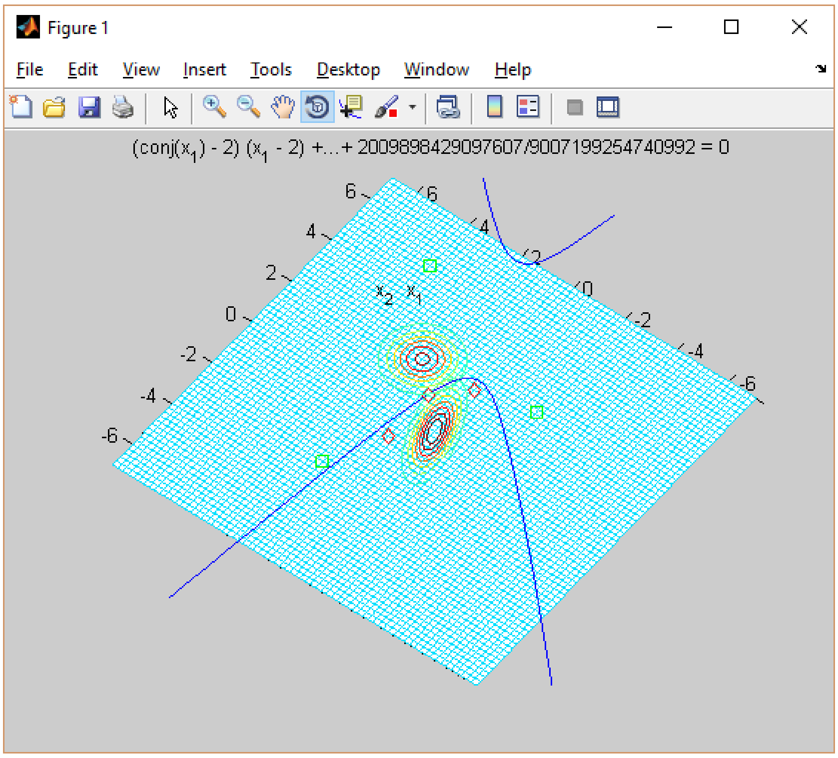

Figure 6: Rotated Output after running the above code

RESULT ANALYSIS

From figure 2,3 and 4 we see that when we changed only the mean, the distribution only shifted in that position resulting in 5 samples to belong in class 1 and one sample in class 2. Closer the two mean comes, more the error rate. Again when only sigma was changed, the distribution changed both remaining in their previous means. And when both were changed, both the position and distribution changed causing 3 sample in Class 1 and 3 in Class2.

CONCLUSION

In this experiment we have implemented the minimum error rate classifier and saw the effect of changing mean and sigma values.